Day 18 | Monday

18

November 30, 2026

Area and material coverage

12:45 PM–2:00 PM • Math • 75 min

Composite areas, openings and surface coverage

2:15 PM–3:30 PM • Math • 75 min

Week 6 • 2 scheduled blocks • 150 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

12:45 PM–2:00 PM • 75 min

Area and material coverage

12:45 PM–12:50 PM	**Start**	5
12:50 PM–12:55 PM	**Recall**	5
12:55 PM–1:10 PM	**Teach**	15
1:10 PM–1:25 PM	**Together**	15
1:25 PM–1:45 PM	**Practise**	20
1:45 PM–1:55 PM	**Check**	10
1:55 PM–2:00 PM	**Wrap up**	5

Math • Math Week 6, Lesson 1

2:15 PM–3:30 PM • 75 min

Composite areas, openings and surface coverage

2:15 PM–2:20 PM	**Start**	5
2:20 PM–2:25 PM	**Recall**	5
2:25 PM–2:40 PM	**Teach**	15
2:40 PM–2:55 PM	**Together**	15
2:55 PM–3:15 PM	**Practise**	20
3:15 PM–3:25 PM	**Check**	10
3:25 PM–3:30 PM	**Wrap up**	5

Math • Math Week 6, Lesson 2

Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time

Area and material coverage

Today’s goal: Calculate rectangle and triangle areas and distinguish linear units from square units.

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Exercise Book 10A Q1–8 (p.93); 10B Q1,4 (p.95)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Recall and reconnect

Explain why 1 m² is not 1,000 mm²; retrieve a rectangle perimeter and name its different unit.

Attempt the retrieval task independently; explain the method before checking.

Resources: Exercise Book 10A Q1–8 (p.93); 10B Q1,4 (p.95)

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

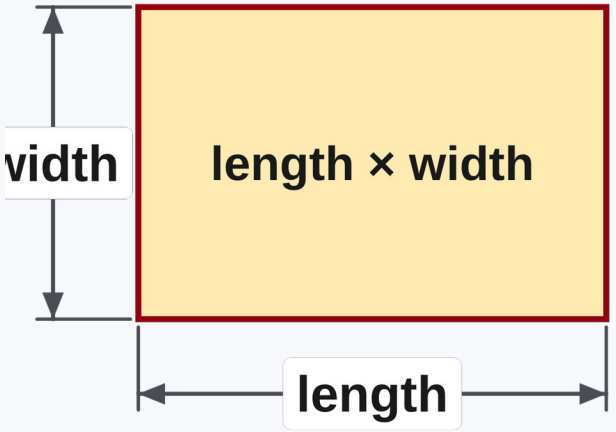
Measure coverage in squares

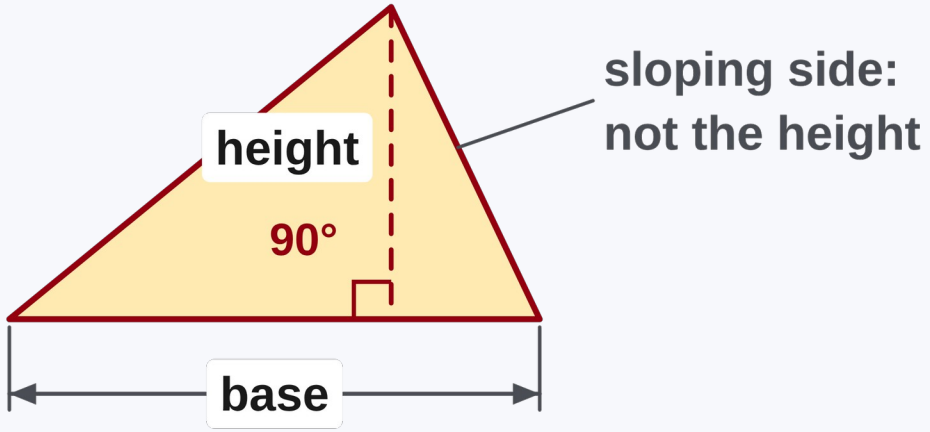
1 Rectangle area = length × width.

2 Triangle area = ½ × base × perpendicular height.

3 The height meets the base at 90°; it is not usually the sloping side.

4 Area answers use square units such as mm² or m².

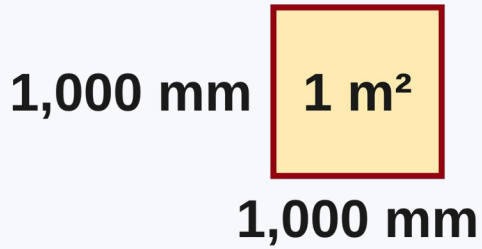




Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Convert areas and count sheets

1. A square conversion applies the length factor twice.
2. Because 1 m = 1,000 mm, 1 m² = 1,000,000 mm².
3. Sheet count = surface area ÷ one-sheet area in matching units.
4. Keep the exercise’s no-waste assumption explicit; round purchases as required.



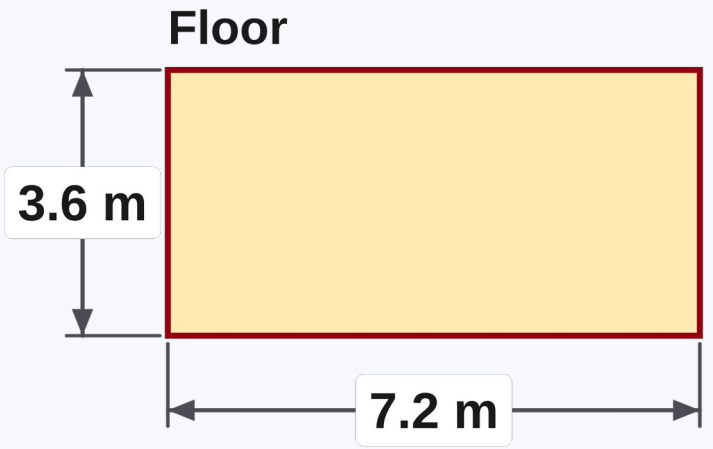
1 m = 1,000 mm

1 m² = 1,000 mm × 1,000 mm = 1,000,000 mm²

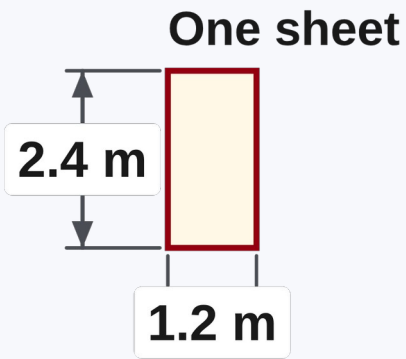
Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Turn floor area into a sheet quantity

Floor: 7.2 m × 3.6 m = 25.92 m²
One sheet: 1.2 m × 2.4 m = 2.88 m²
No-waste quantity: 25.92 ÷ 2.88 = 9 sheets



What the result means • This is a no-waste classroom calculation.
An area quotient is not automatically a workable cutting plan.
Keep the sheet dimensions and layout assumptions visible.



Together • Work 10A Q2–4,7–8. Label perpendicular heights.
Shape Canvas: compare a rectangle and a right triangle with the same base and height. Predict why the triangle has half the area.

Exercise Book 10B Q1 p.95 • Numeracy Hub → Volume and Layout → Area and perimeter

Textbook | page 244

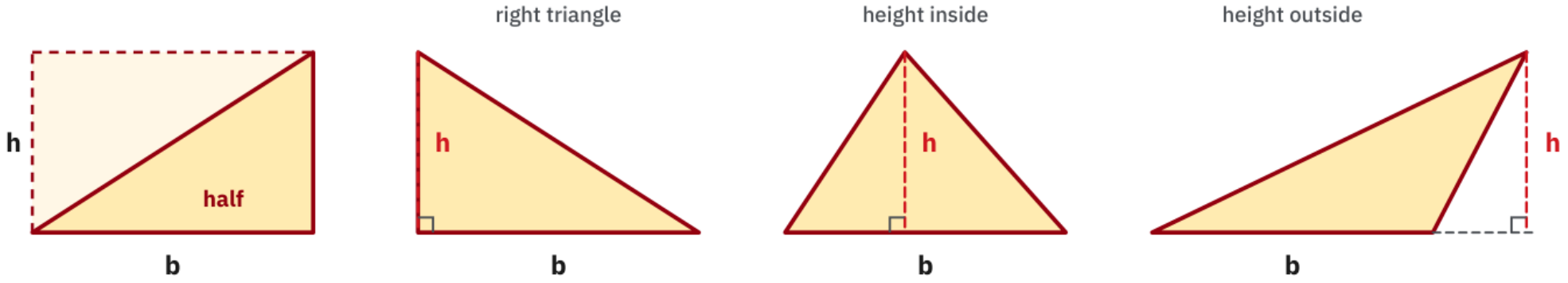


Figure 10.3 Left: a rectangle cut on its diagonal makes two equal triangles. Right: three triangles with the same base b and the same height h have the same area.

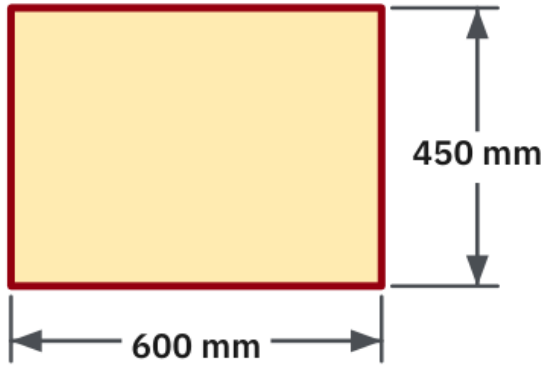
Textbook p.244 • Figure 10.3 — Equal base and perpendicular height

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Textbook | page 243

Example 10.1 Area of a base plate

A machine base plate is 600 mm long and 450 mm wide. Find its area in mm² and in m².



SOLUTION

1. $A = l \times w = 600 \times 450 = 270{,}000$ mm².
2. To change mm² to m², divide by 1,000,000:
   $270{,}000 \div 1{,}000{,}000 = 0.27$ m².
3. Check: in metres the plate is 0.6 m by 0.45 m, and
   $0.6 \times 0.45 = 0.27$ m².

Textbook p.243 • Example 10.1 — Rectangle area and squared unit conversion

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Spot the mistake. Explain the correction.

Can an area in m² be converted to mm² by multiplying by 1,000?

Discuss the method before checking the answer.

No. Both dimensions change by 1,000, so the area factor is 1,000 squared.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Work together

Work Exercise Book 10A Q2–4,7–8; learners label perpendicular height, not the sloping side.

Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.

Resources: Exercise Book 10A Q1–8 (p.93); 10B Q1,4 (p.95)

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Guided practice | 10A Q2

Write 1,500,000 mm² in m².

Work with a partner. Take turns explaining the next step.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Compare your method | 10A Q2

Write 1,500,000 mm² in m².

CHECK TOGETHER

1.5 m²

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Your scheduled practice

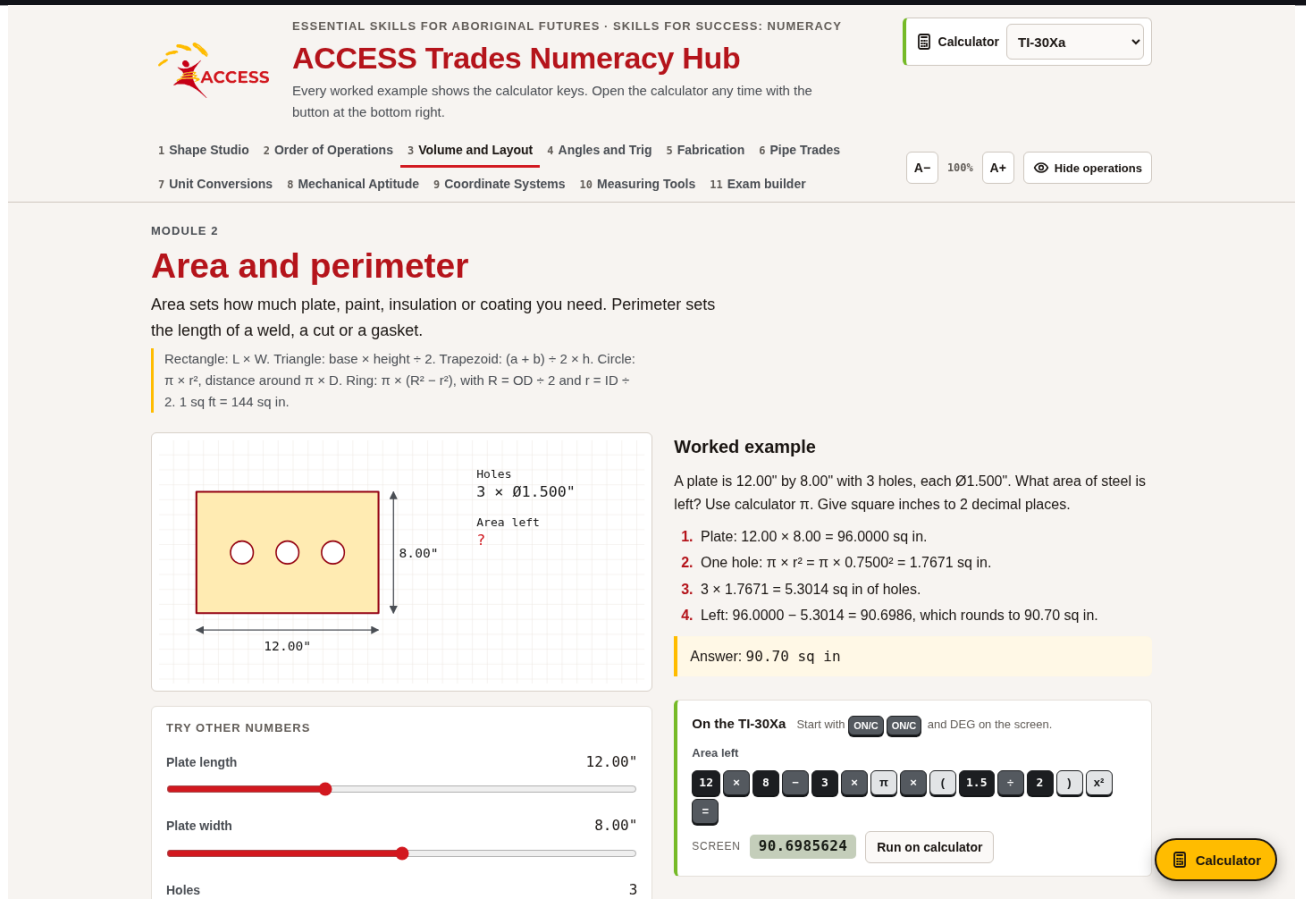
Complete Exercise Book 10A Q5–6 and 10B Q1,4. On Shape Canvas, compare the two shapes' areas. Keep the sheet count's no-waste assumption explicit.

Solve on paper first; check with tools; correct one error and explain the change.

Resources: Exercise Book 10A Q1–8 (p.93); 10B Q1,4 (p.95)

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Open Numeracy Hub



Navigation example. Use the values and units in the assigned task.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Practise in Numeracy Hub and Shape Canvas

1. Create a rectangle and right triangle with equal base and height in Shape Canvas.

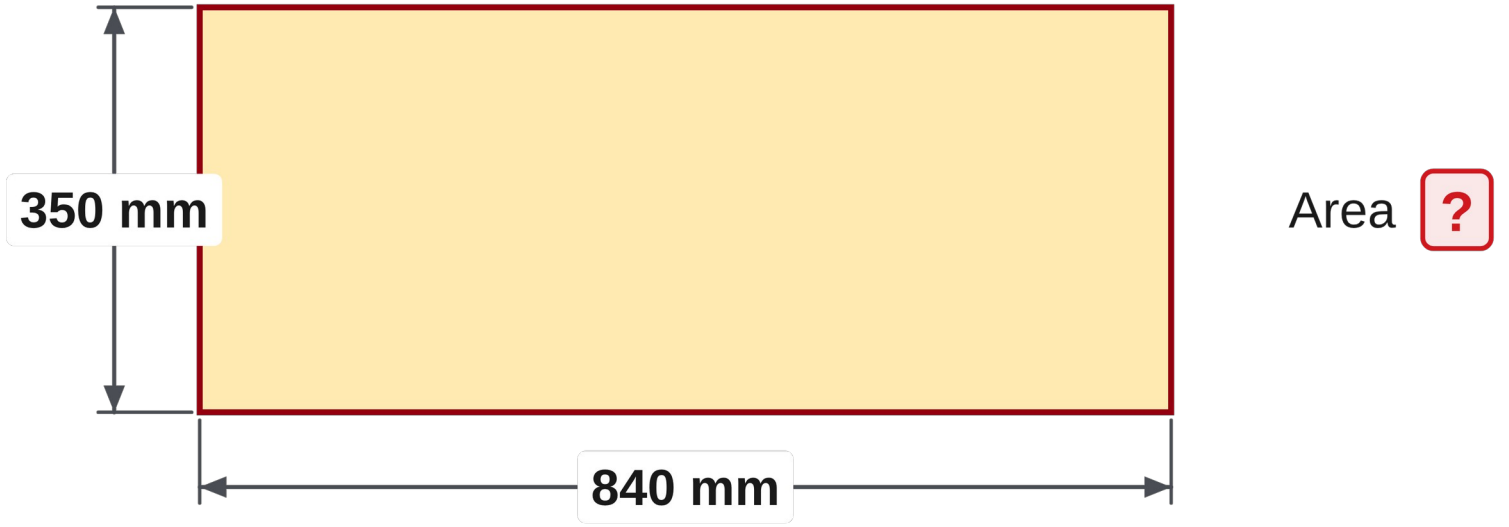
2. Calculate both areas on paper; explain why the triangle covers half as much.

3. Check the area method in Numeracy Hub Volume and Layout → Area and perimeter.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Practise from the Exercise Book | 10A Q5

Find the area of a rectangle 840 mm by 350 mm.

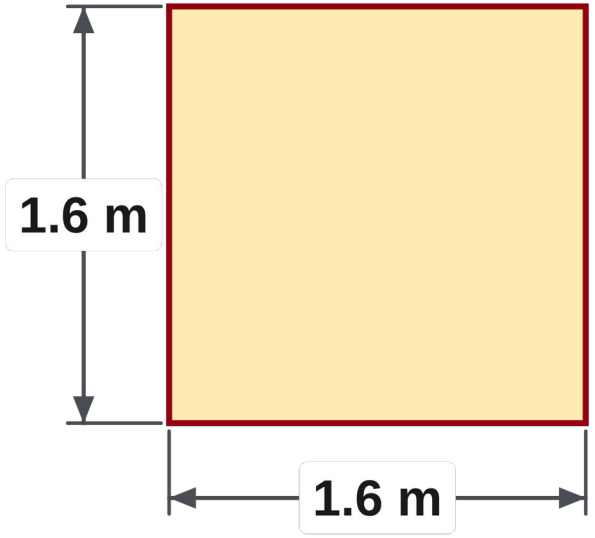


Write the operation, working and units before checking.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Practise from the Exercise Book | 10A Q6

Find the area of a square with 1.6 m sides.



Area ?

Write the operation, working and units before checking.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Check independently

Exercise Book 10B Q1; show floor area, one-sheet area and sheet quantity.

Solve without displayed answers; include working, units and final rounding.

Resources: Exercise Book 10A Q1–8 (p.93); 10B Q1,4 (p.95)

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

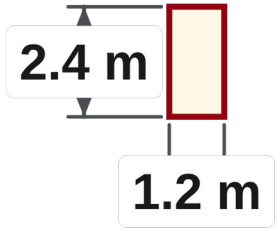
Independent check | 10B Q1

A mezzanine floor is 7.2 m by 3.6 m. It is covered with checker plate sheets that are 1.2 m by 2.4 m, laid edge to edge with no waste. Find the floor area and the number of sheets.

Floor



One sheet



Floor area ?

Sheets ?

Answer independently. Keep worked solutions closed.

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

Name the math and one other essential skill used; record a next practice step.

Resources: Exercise Book 10A Q1–8 (p.93); 10B Q1,4 (p.95)

Textbook sections 10.1–10.3 pp.242–245; sections 7.4 pp.170–172

Break

2:00 PM–2:15 PM

Return ready for: Composite areas, openings and surface coverage

This break is excluded from teaching minutes.

Composite areas, openings and surface coverage

Today’s goal: Break a composite shape into known parts and subtract openings before estimating coverage.

Describe a prior use of the skill; identify the given and unknown quantities.

Textbook sections 10.4–10.7 pp.246–254

Exercise Book 10A Q10–12 (p.94); 10B Q2 (p.95), Q11–12 (p.96); 10C Q4 (p.97)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook sections 10.4–10.7 pp.246–254

Recall and reconnect

Show a panel with a hole; ask which area is added and which is subtracted.

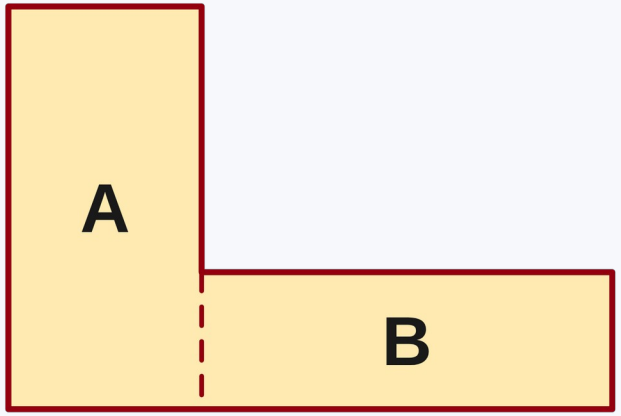
Attempt the retrieval task independently; explain the method before checking.

Resources: Exercise Book 10A Q10–12 (p.94); 10B Q2 (p.95), Q11–12 (p.96); 10C Q4 (p.97)

Textbook sections 10.4–10.7 pp.246–254

Build a composite area from known pieces

1. Split the outline into non-overlapping shapes with known area formulas.
2. Label the required lengths and perpendicular heights.
3. Add the separate areas once each.
4. For a trapezoid, multiply the average parallel-side length by its height.



split on the dashed line

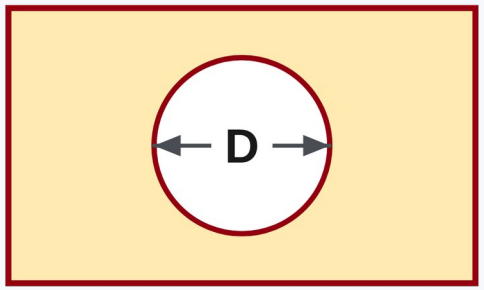
area = A + B

each part added once

Textbook sections 10.4–10.7 pp.246–254

Subtract openings before estimating coverage

1. Net area = outer area − opening area.
2. For a circular opening, use $A = \pi r^2$ with half the diameter as r.
3. Keep all areas in the same square unit.
4. Use net area and the stated coverage to calculate whole tins to buy.



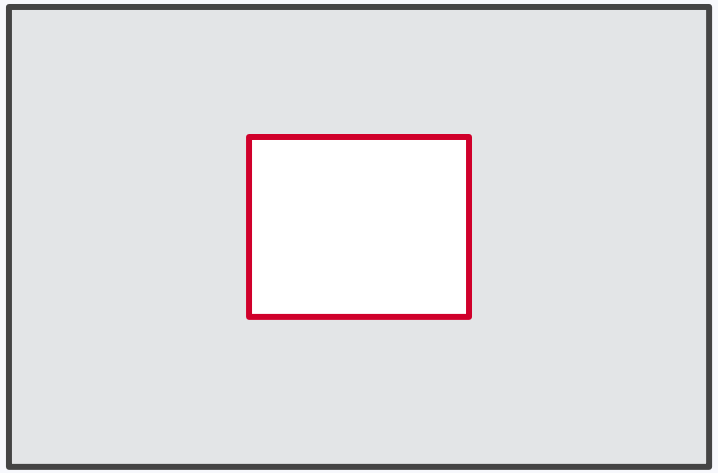
net area = outer area − opening

circle opening: r = D ÷ 2

Textbook sections 10.4–10.7 pp.246–254

Subtract the opening, not the outside

Outer: 1,200 × 800 mm



Opening: 300 × 200 mm

Outer area = 960,000 mm²
Opening area = 60,000 mm²
Net = 900,000 mm² = 0.9 m²

Check: 900,000 + 60,000
= 960,000 mm².
Keep all dimensions in one unit.

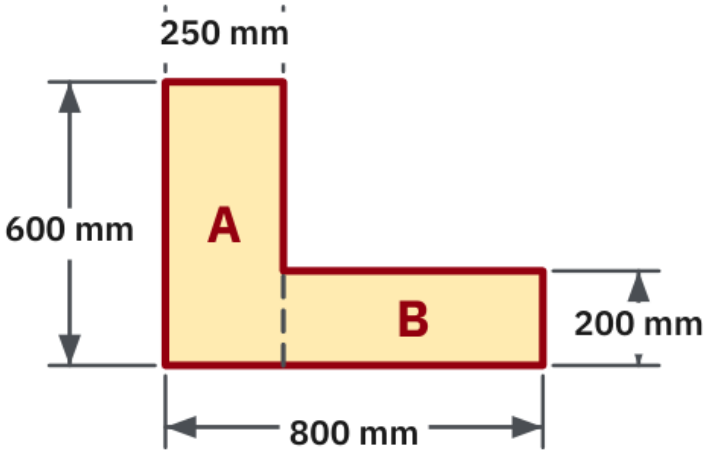
Together • 10A Q10–12 and 10C Q4. Annotate non-overlapping parts.
Use a Shape Canvas sketch only after choosing the add/subtract steps.

Exercise Book 10B Q11 p.96 • Schematic, not to scale • Surface-net preview is optional after core work

Textbook | page 251

Example 10.7 An L-shaped plate

Find the area of the L-shaped plate.



SOLUTION

1. Split on the dashed line into A (the upright) and B (the foot).
2. A: 250 × 600 = 150,000 mm².
3. B is 800 − 250 = 550 mm long: 550 × 200 = 110,000 mm².
4. Add: 150,000 + 110,000 = 260,000 mm².

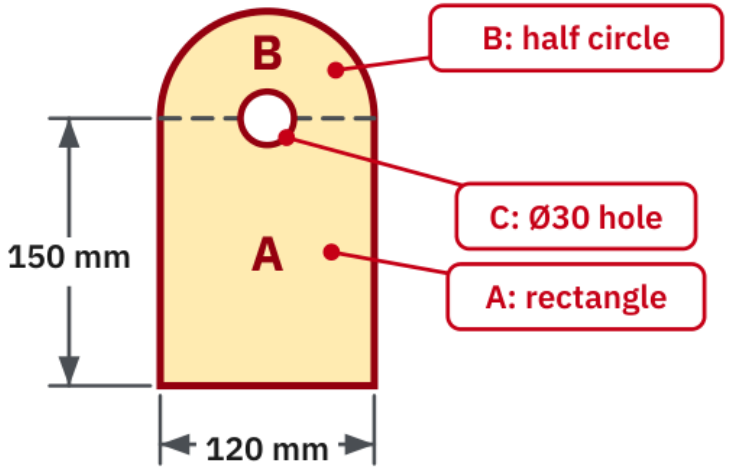
Textbook p.251 • Example 10.7 — Split an L-shaped plate into non-overlapping rectangles

Textbook sections 10.4–10.7 pp.246–254

Textbook | page 252

Example 10.8 A lifting lug

A lifting lug is a 120 mm by 150 mm rectangle with a half circle on top. A 30 mm diameter hole is centred where the half circle meets the rectangle. Find the area of the lug to two decimal places.



SOLUTION

1. A: 120 × 150 = 18,000 mm².
2. B: the half circle is 120 mm across, so r = 60 mm. π × 60² ÷ 2 = 5,654.87 mm².
3. C: r = 15 mm. π × 15² = 706.86 mm².
4. A + B − C = 22,948.01 mm².

Textbook p.252 • Example 10.8 — Add the shapes and subtract the opening

Textbook sections 10.4–10.7 pp.246–254

Spot the mistake. Explain the correction.

A panel has a circular opening labelled by diameter. Can that diameter go directly into πr^2?

Discuss the method before checking the answer.

No. Divide the diameter by two to obtain the radius, then square the radius.

Textbook sections 10.4–10.7 pp.246–254

Work together

Work Exercise Book 10A Q10–12 and Exercise Book 10C Q4. Partners annotate non-overlapping pieces.

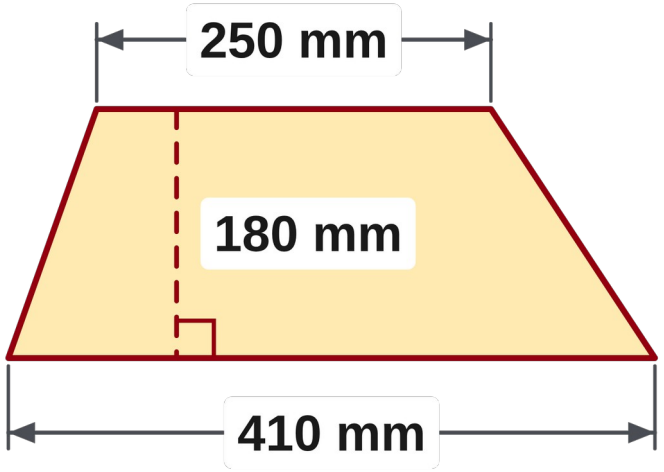
Work in pairs, alternating solver and checker. Write a diagram, formula, units and estimate.

Resources: Exercise Book 10A Q10–12 (p.94); 10B Q2 (p.95), Q11–12 (p.96); 10C Q4 (p.97)

Textbook sections 10.4–10.7 pp.246–254

Guided practice | 10A Q10

Trapezoid: parallel sides 250 mm and 410 mm, height 180 mm. Find the area.



Area ?

Work with a partner. Take turns explaining the next step.

Textbook sections 10.4–10.7 pp.246–254

Compare your method | 10A Q10

Trapezoid: parallel sides 250 mm and 410 mm, height 180 mm. Find the area.

CHECK TOGETHER

59,400 mm²

Textbook sections 10.4–10.7 pp.246–254

Your scheduled practice

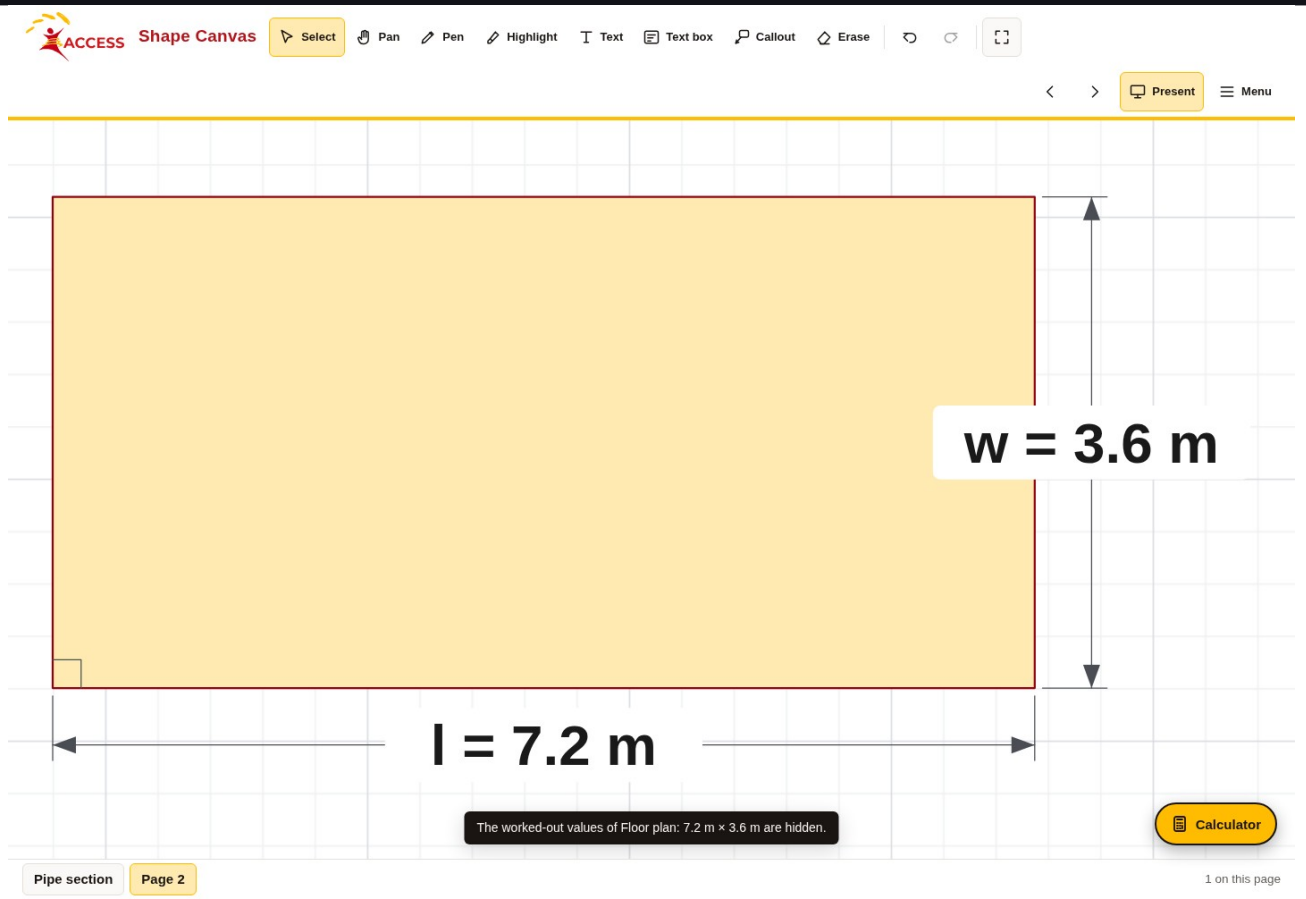
Complete Exercise Book 10B Q2,11–12. Build a Shape Canvas sketch and label areas; use the net preview only if core subtraction and coverage are secure.

Solve on paper first; check with tools; correct one error and explain the change.

Resources: Exercise Book 10A Q10–12 (p.94); 10B Q2 (p.95), Q11–12 (p.96); 10C Q4 (p.97)

Textbook sections 10.4–10.7 pp.246–254

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook sections 10.4–10.7 pp.246–254

Practise in Numeracy Hub and Shape Canvas

1. Use two labelled shapes in Shape Canvas to represent the outer rectangle and opening.

2. Compute the separate areas and net area on paper before comparing the labels.

3. Only after core coverage work is secure, inspect the optional Shape Studio solid/net preview.

Textbook sections 10.4–10.7 pp.246–254

Practise from the Exercise Book | 10B Q2

Four plates, each 2.5 m by 1.6 m, are primed on both sides. The primer label says 1 L covers 7 m². Primer is sold in 1 L tins. How many tins do you buy? Give the litres to two decimal places.



× 4 plates

both sides

Primer: 1 L covers 7 m²

Sold in 1 L tins

Litres ?

Tins ?

Write the operation, working and units before checking.

Textbook sections 10.4–10.7 pp.246–254

Practise from the Exercise Book | 10B Q11

A 300 mm by 200 mm access hole is cut in a 1,200 mm by 800 mm cover plate. Find the area of plate left in mm² and in m².

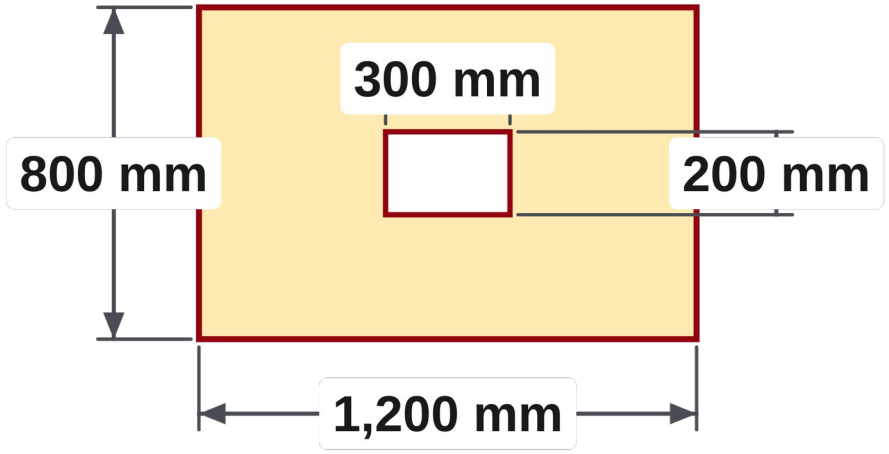


Plate left (mm² and m²) ?

Write the operation, working and units before checking.

Textbook sections 10.4–10.7 pp.246–254

Check independently

Exercise Book 10B Q11 and Q2; report net area and whole tins to buy.

Solve without displayed answers; include working, units and final rounding.

Resources: Exercise Book 10A Q10–12 (p.94); 10B Q2 (p.95), Q11–12 (p.96); 10C Q4 (p.97)

Textbook sections 10.4–10.7 pp.246–254

Independent check | 10B Q11

A 300 mm by 200 mm access hole is cut in a 1,200 mm by 800 mm cover plate. Find the area of plate left in mm² and in m².

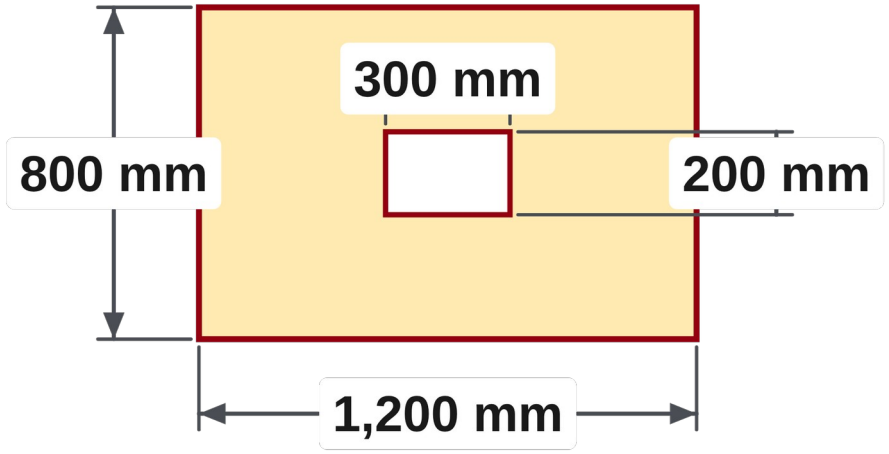


Plate left (mm² and m²) ?

Answer independently. Keep worked solutions closed.

Textbook sections 10.4–10.7 pp.246–254

Independent check | 10B Q2

Four plates, each 2.5 m by 1.6 m, are primed on both sides. The primer label says 1 L covers 7 m². Primer is sold in 1 L tins. How many tins do you buy? Give the litres to two decimal places.



× 4 plates

both sides

Primer: 1 L covers 7 m²

Sold in 1 L tins

Litres ?

Tins ?

Answer independently. Keep worked solutions closed.

Textbook sections 10.4–10.7 pp.246–254

Wrap up and choose the next step

Ask: What changed in your understanding? Where will you use it? What should you check next time?

Name the math and one other essential skill used; record a next practice step.

Resources: Exercise Book 10A Q10–12 (p.94); 10B Q2 (p.95), Q11–12 (p.96); 10C Q4 (p.97)

Textbook sections 10.4–10.7 pp.246–254